AI for Economic Research: Dynamic Models, Language, and Agents
Lecture 2.2: Language, Generation, and Agency

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Topic 2.2 Agenda

1. **What Is Intelligence?** — the cognitive debate
2. **From NLP to LLMs to Agents** — the generative stack
3. **How People Use Generative AI**
4. **Caution: Why Generative AI Still Fails** — hallucination & deepfakes

Arc: T2.1 Predictive stack (and where it fails) → T2.2 Generative & agentic (and where it fails)

What Is Intelligence? The Cognitive Debate

What is Intelligence?

- **Problem-Solving**: The capacity to identify problems and devise effective solutions.
- **Adaptability**: The ability to adjust behavior in response to changes in the environment.
- **Learning**: Acquiring new knowledge or skills through experience and education.
- **Reasoning**: Drawing conclusions based on available information, including logical reasoning and inference.
- **Classification and Generalization:**
  - Rosch (1975): people categorize objects by comparing them to an idealized prototype.
  - Chomsky (1965, *Aspects*): gives the modern generative formulation of universal grammar (reviving a much older idea) — all human languages share a common underlying structure.
  - *AI tries to learn these same cognitive capabilities from data.*

The Cognitive Debate: Stochastic Parrots vs. Reasoners

- **The Stochastic Parrot Hypothesis (Bender et al., 2021, FAccT '21):**
  - LLMs produce plausible text without communicative intent or world models.
  - Valid simulators of language — but not cognitive agents.
- **Chomsky's Critique:**
  - LLMs excel at **description and prediction** but fail at **explanation**.
  - True intelligence requires causal reasoning; LLMs rely on probabilistic correlation.

The dividing question: is fluent language *evidence of understanding*, or a convincing imitation of it? Economists face the same question whenever a model "fits" the data without explaining it.

The Reasoning Gap: Induction vs. Abduction

- **Where AI is strong — inductive reasoning (归纳推理):** generalizing from massive data distributions (more examples $\Rightarrow$ better).
- **Where humans are strong — abductive reasoning (溯因推理):** inferring the *best explanation* from limited evidence.
- **The benchmark gap (Chollet, 2019; ARC Prize 2024):** LLMs dominate semantic benchmarks (MMLU) yet struggle with the **Abstraction and Reasoning Corpus (ARC)**, which tests on-the-fly adaptation to *novel* abstract rules.

Bottom line for economists: lean on AI where inductive pattern recognition dominates; stay skeptical where novel *causal judgment* matters.

AI vs. Human Intelligence: Strengths and Limits

	Human Intelligence	**AI (Current)**
Pattern matching	Good	Superhuman
Novel causal reasoning	Strong	Weak
Explanation	Strong	Weak
Common sense	Strong	Improving
Replication cost	20+ years	Near-zero
Parallelism	One body	Millions of instances
Energy	$\sim$20W	Kilowatts (at scale)
Domain range	Broad	Narrow $\rightarrow$ General
Context memory	Long-term	Context window

From NLP to LLMs to Agents

Natural Language Processing: From Bag-of-Words to Transformers

- **Generation 1 — Bag-of-Words:**
  - Represent text as a vector of word counts, ignoring grammar and order
  - TF-IDF weights words by frequency in document vs. corpus
  - Application: basic document classification, keyword search
- **Generation 2 — Word Embeddings (Word2Vec, GloVe, 2013–2014):**
  - Map words to continuous vectors capturing semantic similarity
  - "king $-$ man $+$ woman $\approx$ queen"
  - Application: measuring sentiment, ideology shifts in text
- **Generation 3 — Transformers and LLMs (2017–present):**
  - Self-attention: weight each word by its relevance to every other word in the sequence
  - Pre-trained on internet-scale data; fine-tuned or prompted for specific tasks
  - Unified architecture handles text, code, images, time series

Large Language Models: Predicting the Next Word

- **One simple objective.** An LLM repeatedly predicts the *next token* given everything before it — and that single trick, at enormous scale, produces fluent, useful text:

$$P(w_1, w_2, \ldots, w_T) = \prod_{t=1}^{T} P(w_t \mid w_1, \ldots, w_{t-1})$$

(just next-word prediction, over and over.)

- **What it is built from** (mechanics in Lec. 7):
  - *Tokenization:* break text into subword pieces.
  - *Embeddings:* map tokens to vectors that capture meaning.
  - *Self-attention (the Transformer):* let each word weigh every other word, capturing long-range context.

LLMs: The Knowledge-Cutoff Problem

- An LLM is trained on a *static snapshot* of data: it does not "know" about events after its training cutoff, and cannot draw on a source it never saw.
- **The fix — Retrieval-Augmented Generation (RAG):** fetch relevant, up-to-date documents at query time and condition the model on them (Lec. 8).

Preview: full treatment in Lec. 7 (LLMs) and Lec. 8 (RAG) — how to feed *your* data and sources to an LLM.

Reasoning & Agentic AI (Preview)

- **The Reasoning & Agentic Era (2024–):**
  - *Reasoning models* (OpenAI o1/o3, Claude thinking, DeepSeek-R1) trade inference compute for chains of thought — a partial answer to Chollet's ARC critique (Chollet, 2019; ARC Prize 2024).
  - *Open-source frontier* (Llama 3+, DeepSeek, Qwen) closes the gap with proprietary models; lowers the cost of academic experimentation by orders of magnitude.
  - *Agentic AI:* models act autonomously, plan multi-step tasks, write and execute code, and drive research workflows.

- **Why preview it here?** Generation (this deck) is the substrate; *agency* is generation wired into a loop with tools, memory, and goals.

Full treatment in Lec. 9 (Agentic AI); for research workflows that string these together, see the case studies in Lec. 10.

Technique Taxonomy: The Five Families

Family	What it is	Where in the course
Machine Learning	learn $f : x \to y$ from data (sup./unsup.)	Lec. 3
Deep Learning	neural nets as universal approximators	Lec. 3–4
Reinforcement Learning	sequential decisions = Bellman equation	Lec. 5–6
Large Language Models	transformers over token sequences; generation	Lec. 7–8
Agentic AI	LLMs + tools + planning loops	Lec. 9–10

One lineage: **ML** learns patterns → **DL** scales them → **RL** acts on them → **LLMs** generate language → **agents** act through language. The rest of the course walks this ladder.

How People Use Generative AI

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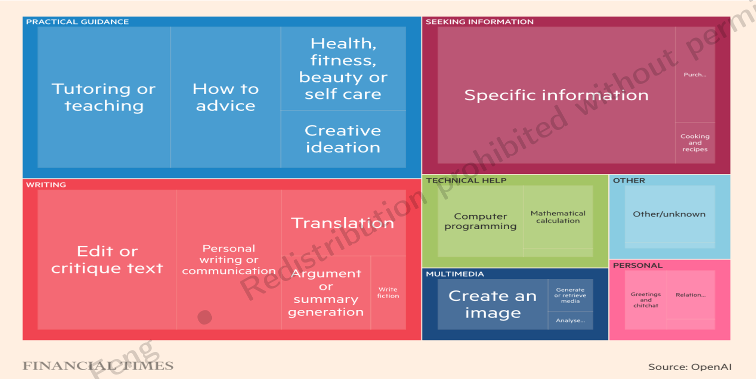
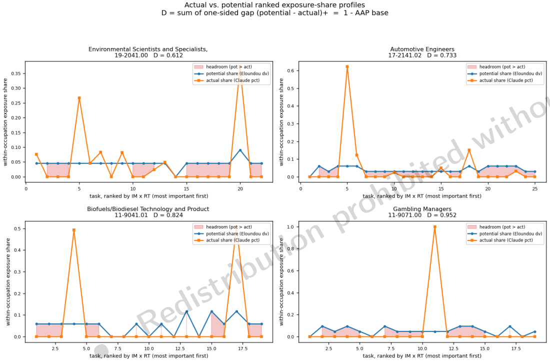


Consumer usage is dominated by *generative* tasks — writing, coding, tutoring, and open-ended Q&A. But *who* uses it, and for *which* tasks, varies enormously across the economy — something we can now measure directly. →

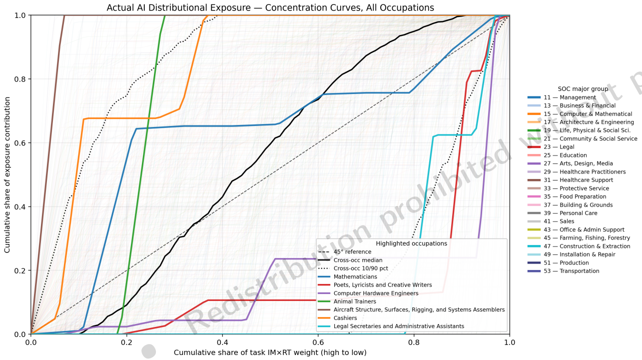
How AI Is *Actually* Used Within an Occupation



Each panel is one occupation; its tasks run along the x-axis, *most important first*. **Orange = where AI is actually used**; **blue = where it *could* be**. Actual use *spikes on one or two tasks* and leaves the rest untouched — and the pattern differs sharply by occupation.

*Source: author's analysis — Anthropic Economic Index (Claude usage) × O*NET importance × Eloundou et al. (2023).*

Zooming Out: AI-Usage Concentration Across 800+ Occupations



Each faint line is one occupation's *concentration curve* — how fast AI usage accumulates as tasks are added most-to least-important. *Above* the 45° line = usage on **core** tasks (Cashiers, Mathematicians); *below* = on **peripheral** tasks. AI's footprint is highly uneven.

*Source: author's analysis — Anthropic Economic Index (Claude usage), 833 O*NET occupations.*

Caution: Why Generative AI Still Fails

Hallucination: Automating Plausibility, Not Truth

- LLMs are trained to produce *plausible* text, not *true* statements: there is **no source of truth during training**. Fabrication is therefore *inherent* — not a bug a future patch removes.
- **Anchor (Mata v. Avianca, 2023):** a New York lawyer filed a ChatGPT-written brief that cited **entirely fabricated cases and invented judicial opinions**; asked whether they were real, the model insisted they were. The lawyer was sanctioned.
- **It is systematic, not a one-off:**
  - **CNET:** of 77 AI-written articles, **41 contained factual errors** — despite a "fact-checked" label.
  - *Defamation by chatbot:* models have fabricated news stories accusing real people of crimes they never committed.

Implication: generative AI is *unreliable for any high-stakes factual task* unless every claim is independently verified.

Source for this section's cases: Narayanan & Kapoor, *AI Snake Oil* (2024).

Deepfakes & Malicious Use: Capability Turned Against People

- The same generative capability produces realistic **voice, image, and video** from seconds of source material.
- **Fraud (2019):** a UK energy firm's managing director wired $\approx$ EUR 220,000 to a fake supplier after a **voice-cloned call** impersonating his boss — one of the first confirmed AI-enabled scams.
- **Non-consensual harm:** a single Telegram bot generated **100,000+ fabricated intimate images** of real women without their consent; the *large majority* of harmful deepfakes target women this way ("image-based abuse").
- **The "liar's dividend":** once *anything* could be fake, real evidence can be waved away as fake too — truth itself loses its anchor.

Implication: *capability* $\neq$ *safety* — a more capable generator is also a more capable weapon.

Open Weights and the Safety Frontier: The DeepSeek Case

A genuine capability leap — and a live case study in *capability* $\neq$ *safety*.

- **The upside.** DeepSeek's open-weight models (V3, R1; 2025) matched frontier reasoning at a fraction of the cost — exactly the "open-source frontier" that lowers the cost of academic experimentation.
- **The catch — weak guardrails.** Independent red teams bypassed R1's safety filters with ease: Cisco reported a ~*100% attack-success rate* on standard jailbreaks; others elicited malware, ransomware, and weapon-making instructions.
- **Open weights make safety *optional*.** Anyone can download the weights and *fine-tune the guardrails away*; safety that lives only inside the released model is not a real barrier ("illusory safety").
- **Governance fallout.** Data stored under PRC jurisdiction, an exposed user-log database, and topic censorship led *many governments* to ban it from official devices (Italy, Australia, South Korea, Taiwan, U.S. federal/state agencies).

For researchers: open models are invaluable for reproducibility and cost — but *openness shifts the safety burden onto you*, the deployer. Capability, cost, and safety are *separate* axes. *(Status as of 2026; red-team findings from Cisco, KELA, Palo Alto Unit 42.)*

The Practitioner's Discipline: What Task? What Data? What Claim?

- **Reliability $\neq$ capability.** The two failure modes differ in kind:
  - *Predictive failure* (T2.1): the model **can't** — the future isn't in the data (COMPAS, Fragile Families).
  - *Generative failure* (this deck): the model **can, but unreliably or harmfully** (hallucination, deepfakes).

- **The habit that guards against both** — before trusting any AI output, ask:
  1. **What task** is this, really? (a prediction? a generation? a decision?)
  2. **What data** did it learn from, and does my setting match?
  3. **What claim** am I making on its output, and how would I verify it?

AI is powerful *but bounded*. We return to this discipline in **Lec. 7 T3b** (LLM limitations) and turn it into a publication-grade **audit checklist in Lec. 9 T5**.

The Same Discipline, for Economists

The three questions, sharpened for research use:

1. **What task?** Is the AI output a *measurement* (a new variable), a *prediction*, or a *decision rule*? Each enters a paper differently — and a measured variable is usually the most defensible.

2. **What data?** Does the training distribution match *your* sample and period? Watch for *look-ahead / leakage* (a text model that already "knows" the outcome) and *regime shifts* — the Lucas-critique problem from T2.1.

3. **What claim?** If an AI-generated variable sits on the *right-hand side*, it carries *measurement error* — attenuation and generated-regressor problems. If it is the *outcome*, validate it against ground truth on a held-out, human-coded sample.

Rule of thumb: treat an AI output as an *estimated input*, never as ground truth — report the model, version, and prompt for *reproducibility*, and robustness-check the AI step as you would any first-stage estimate. (Toolkit: Lec. 7–9.)

What Lec. 3 Delivers

- **Lec. 3 T1 — Why ML for macro:** unstructured data (text, satellite, panels), curse of dimensionality, pattern recognition as motivation.
- **Lec. 3 T2 — Neural networks:** architecture, activations (ReLU, sigmoid, tanh), universal approximation (Cybenko 1989, Hornik 1991).
- **Lec. 3 T3 — Training:** forward/back propagation, gradient descent + AdamW, mini-batches, regularization (dropout, weight decay).
- **Lec. 3 T4 — Specialized architectures:** ICNN, monotone NN, PICNN with macro shape restrictions.

Lec. 3 is the toolkit; Lec. 4 turns it on the optimal-growth Bellman equation.